# Designing Safer And Better ELECTRIC CAR BATTERIES

As technology advances, batteries evolve. From aerogels to CTP designs, what are the latest breakthroughs in battery safety and performance?

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Thermal runaway in electric vehicle (EV) batteries is a critical safety concern that demands careful attention. This phenomenon can occur due to various factors, such as overheating or internal and external short circuits. When thermal runaway occurs, it triggers a rapid increase in temperature and the release of volatile gases, potentially leading to a dangerous cascade effect if it spreads to other cells within the battery pack. Such incidents can result in temperatures exceeding a thousand degrees, and the highly volatile gases can ignite upon contact with high voltages, posing significant risks.

The fire risks associated with EVs differ markedly from those in combustion engine vehicles, where flammable liquids like petrol and diesel are the primary concern. In EVs, energy storage devices contain materials such as lithium, necessitating unique fire safety considerations. The potential for gas venting in confined spaces, such as garages, can create explosive conditions, and EV fires may take longer to extinguish. Moreover, the evolving nature of regulations for EVs, which often lag technological advancements, adds complexity to safety measures.

As demand for EVs continues to rise and safety regulations become more stringent, the need for advanced fire protection materials becomes increasingly